

A8 · TCP on a Noisy Link

CS425 Computer Networks · instructor led · every command runs on Onyx (ssh onyx)

TURN THIS IN ON PAPER

before you leave the room
graded pass/fail

Names _____ Date _____

Every round: **predict** first, in ink, then **run** the command with the projector, then **check**. A wrong prediction costs nothing. A prediction written afterwards is not a prediction.

```
wget -q https://shanepanter.com/cs425/activities/tcpmeter.py
unshare -rn bash                # a private network stack, where you are root
ip link set lo up mtu 1500
tc qdisc add dev lo root netem delay 25ms rate 20mbit
```

ROUND 1 · BUILD A SLOW LINK `ping -c 3 127.0.0.1`

QUESTION	ANSWER
RTT to 127.0.0.1 before tc	
Predict: RTT after netem delay 25ms	
RTT ping reported	
Transmission delay of one 1,500 byte frame at 20 Mbit/s	

Q1. Why is the RTT 50 ms and not 25? How many times does each ping cross lo?

ROUND 2 · A CLEAN LINK `python3 tcpmeter.py`

QUESTION	ANSWER
Predict: bytes in flight to fill 20 Mbit/s with a 50 ms RTT, and that many 1,448 byte segments	
Average Mbit/s · segments retransmitted	
cwnd and rtt at second 10	

Scratch. Work the bandwidth-delay product here, in bits, then bytes, then segments.

Q2. The window grew far past your prediction. Where are the extra segments? How long does the link take to drain them, and does that match the rtt column?

ROUND 3 · ADD NOISE `tc qdisc change dev lo root netem delay 25ms rate 20mbit loss 2%`

Section 3.7: $\text{throughput} = 1.22 \times \text{MSS} / (\text{RTT} \times \sqrt{L})$, with the MSS in bits and the RTT in seconds.

QUESTION	ANSWER
Predict: average Mbit/s with 2% loss	
The formula's answer, with your working	
Measured average Mbit/s · typical cwnd	

Q3. Nothing on this link was full. Why did TCP give up most of it after losing only 2% of its segments?

ROUND 4 · WHAT 802.11 DOES ABOUT IT

20% of frames are lost on each try, and the access point gives each frame up to 7 tries.

QUESTION	ANSWER
Predict: fraction of frames lost after all 7 tries	
Predict: average tries per frame, and the Mbit/s left for new data	
Predict: TCP's Mbit/s with no retries (TCP sees 20%) · with retries	
Measured: <code>loss 20% · rate 16mbit loss 0.0013%</code>	

Scratch. The loss after 7 tries, the average number of tries, and the formula from round 3 with each loss rate.

Q4. Same radio, same 20% loss. Which layer recovered the loss in each run, and which run is the local recovery fix from section 7.6?

Exit. Two laptops on opposite sides of the room use the same access point and cannot hear each other. Why can 802.11 not detect a collision while sending, the way Ethernet does, and what does it do instead?

Graded pass/fail on something in every box. Hand this in before you leave.